PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Relation between otitis media and sensorineural hearing loss: a systematic review
AUTHORS	Elzinga, Hester; van Oorschot, Hanneke; Stegeman, Inge; Smit, Adriana

VERSION 1 – REVIEW

REVIEWER	Coates, Harvey Princess Margaret Hospital for Children, Perth, Australia, Department of Otolaryngology
REVIEW RETURNED	13-Apr-2021

GENERAL COMMENTS	This review of the potential relationship between CSOM and SNHL has obviously been a monumental but necessary body of work and I congratulate the authors on their diligence in reviewing some 3725 studies.The conclusions are necessarily broad but appropriate and the complexities and limitations of the study well outlined. This paper proposes more longstanding prospective studies with strict inclusion criteria and I agree the need for this is necessary. Specifically, I do not agree with the authors on the statement that in children that there is no correlation between conductive hearing loss and language, speech and scholastic outcomes—in our Australian Aboriginal children ,many of whom have longstanding CSOM there are significant poor speech, language, and educational outcomes— while acknowledging the multiple other comorbidities and social determinants of health plus the dearth of rigid studies showing these outcomes. I applaud the use of the definition of 3 months of discharge for CSOM rather than the two week definition of WHO- we generally use the 6 week definition. I noted typos on line 36, References 5 and 13. Congratulations on this significant contribution to the literature on this subject. Hopefully those of us in countries with a relatively high rate of CSOM in our Indigenous population will take up the challenge to perform prospective studies. In my opinion these studies will be more accurate as historically in Australia some manuals advocated ototoxic ear drops for up to 16 weeks in patients with CSOM.

REVIEWER	Hellström, Sten Karolinska institutet Department of Clinical Sciences Intervention and Technology, Audiology and Neurotology
REVIEW RETURNED	21-Apr-2021

GENERAL COMMENTS	The present manuscript describes a systematic review concerning the relationship between otitis media and sensorineural hearing loss. This is an old and very important question, which of some reason has not been so extensively investigated, and definitely not in some prospective, randomized, controlled studies. In this systematic review the authors identify 4168 papers related to this subject. The search syntax is well described. After removal of duplicates 3725 were screened and 3662 could be excluded just based on titles and abstracts. Of the remaining 63 articles 54 are excluded after full-text analysis. No prognostic studies were included and a meta-analysis could not be performed due to heterogeneity of reported results. Critical scrutinizing of the left 9 paper ranged their quality and only five were rated good, 2 fair and 2 poor. Selection bias was present in all studies. Interestingly, 6 out of the 9 research studies represent other countries than the common continents related to studies on otitis media – US and Europe. The outcome of investigating the 9 identified studies resulted in a conclusion that no correlation could be seen between otitis media and sensorineural hearing loss. A need for future prognostic studies is emphasized. I find this study well designed and performed by a scientists group well experienced in systematic review methods. The results are well described and analysed. Any conclusion regarding a relation between otitis media and sensorineural hearing loss can not be drawn. Nevertheless reports ending in negative results are important and deserve certain attention. It definitely points toward an area which should be investigated in well designed prospective studies. Against the background above I recommend this paper for publication in BMJ Open.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1	**Response**
'This review of the potential relationship between CSOM and SNHL has obviously been a monumental but necessary body of work and I congratulate the authors on their diligence in reviewing some 3725 studies. The conclusions are necessarily broad but appropriate and the complexities and limitations of the study well outlined. This paper proposes more longstanding prospective studies with strict inclusion criteria, and I agree the need for this is necessary.'	We thank the reviewer for acknowledging our work and the need for properly designed prospective studies on this topic.
'Specifically, I do not agree with the authors on the statement that in children that there is no correlation between conductive hearing loss and language, speech and scholastic outcomes—in our Australian Aboriginal children, many of whom have longstanding CSOM there are	We agree with the reviewer that such consequences are well known for longstanding CSOM exposure. In our introduction we report about the situation of transient conductive hearing loss during the acute inflammatory

significant poor speech, language, and educational outcomes— while acknowledging the multiple other comorbidities and social determinants of health plus the dearth of rigid studies showing these outcomes.'	phase. We have altered our sentence to make this distinction clearer for our readers.
'I noted typos on line 36, References 5 and 13.'	We thank the reviewer for addressing these typos and we have corrected them.

Reviewer 2	**Response**
'I find this study well designed and performed by a scientists group well experienced in systematic review methods. The results are well described and analysed. Any conclusion regarding a relation between otitis media and sensorineural hearing loss can not be drawn. Nevertheless reports ending in negative results are important and deserve certain attention. It definitely points toward an area which should be investigated in well designed prospective studies.'	We agree with the reviewer on the relevance of this topic.

VERSION 2 – REVIEW

REVIEWER	Coates, Harvey Princess Margaret Hospital for Children, Perth, Australia, Department of Otolaryngology
REVIEW RETURNED	07-Jul-2021

GENERAL COMMENTS	Thank you for the changes to the original manuscript which clarify my questions. Regarding the introduction I note the older term serous otitis media rather than the current terminology of otitis media with effusion OME. Similarly the term chronic otitis media COM concerns me— do you mean CSOM or chronic OME or both?

VERSION 2 – AUTHOR RESPONSE

Thank you for reviewing our manuscript again. We agree on OME as more up to date terminology and adjusted this. Regarding the second note: we used the terminology CSOM as you suggested, not COM.